

Model Selection Proposal: An Integrated Framework for AI-Powered Direct Indexing and Large Language Models

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Abstract

This document presents the proposed quantitative model direction for the QWIM capstone project, along with a detailed justification for the selection. After systematically reviewing the project's reference literature — with particular attention to Section 1.3.3 (Personalized Investing) and the seminal work by Schroeder and Kronseder (2024) on *AI-Powered Direct Indexing: Exploring Thematic Universes* — and aligning with our team's overall research plan, I have chosen **AI-Powered Direct Indexing via Tracking Error Minimization** as my core quantitative framework and propose to explore its deep integration with Large Language Models (LLMs).

1 Project Understanding and Research Positioning

The central thesis of this project is to investigate personalized portfolio construction for individual investors within a quantitative wealth management framework. The project requires us to focus on mid-term (3–18 months) and long-term (18+ months) asset allocation scenarios, using daily or lower-frequency financial time series data, and to benchmark our approaches against carefully selected reference strategies. What distinguishes this project is the dual requirement that each team member independently implements a quantitative model while collectively exploring the application potential of Large Language Models in personalized investing, specifically satisfying the rigorous requirement of using LLMs to manage investment narratives to convince and impress audiences.

At its core, the research question we must address is: how can quantitative portfolio models be made not only mathematically optimal but genuinely aligned with each investor's idiosyncratic, non-financial values and thematic preferences? Traditional index investing via generic ETFs forces investors to accept rigid, pre-packaged baskets of securities, completely ignoring personal values (e.g., custom ESG criteria, hyper-specific technological trends like “Generative AI Infrastructure,” or personal career-hedging requirements). Direct Indexing solves this by allowing investors to directly own a customized subset of underlying stocks that replicate a benchmark while tilting toward personal themes.

However, translating a client's abstract, qualitative values into a concrete, executable stock universe is a massive bottleneck. This is precisely where the integration with LLMs delivers the greatest value. My research positioning is as follows: **to construct a complete closed-loop system comprising “LLM-driven thematic thesis extraction → Semantic Information Retrieval & Universe Filtering → Tracking Error Convex Optimization → Narrative-based Performance Attribution Reporting,”** and to validate its effectiveness in a personalized single-stock multi-asset framework.

2 Theoretical Advantages of Direct Indexing and Selection Rationale

The selection of AI-Powered Direct Indexing as the core model is grounded in several key considerations:

1. **Perfect Thematic Alignment with ‘Personalized Investing’:** Direct Indexing represents the frontier of hyper-personalization in modern wealth management. Instead of treating personalization as a simple categorical constraint (e.g., “Exclude Tobacco”), this framework treats the user’s specific values as a dynamic vector space that shapes the entire investment universe, natively fulfilling the project’s core philosophical mandate.
2. **Robust and High-Leverage Mathematical Foundation:** The core quantitative math of Direct Indexing rests on **Tracking Error Minimization** or **Mean-Variance Optimization (MVO)**. Under the standard tracking error paradigm, the objective function is a convex quadratic programming problem designed to minimize the variance of the active returns relative to a broad market benchmark (e.g., the *S&P 500*). This mathematical structure is exceptionally stable, thoroughly understood, and guarantees unique global optimal solutions. It completely avoids the convergence failures of deep reinforcement learning or the analytical intractability of complex stochastic differential equations.
3. **Immediate Industry Applicability and High Feasibility:** Direct Indexing is currently one of the fastest-growing segments in Wall Street wealth management (e.g., BlackRock’s acquisition of Aperio, Vanguard’s acquisition of Just Invest). Implementing this model under the guidance of a quantitative specialist from Bank of America ensures our research directly mirrors institutional-grade innovations. Furthermore, because the optimization relies on standard quadratic programming, it can be seamlessly implemented via highly mature, zero-failure Python libraries such as `PyPortfolioOpt` or `cvxpy`, ensuring the 3-person team can deploy it rapidly without encountering code bottlenecks.
4. **Natural Support for Tax-Loss Harvesting and Dynamic Customization:** Because the investor directly owns the underlying equities rather than an ETF share, the portfolio allows for granular, multi-period dynamic rebalancing. Over a mid-to-long-term horizon, securities can be dynamically substituted based on shifting LLM thematic signals or personal tax situations, aligning perfectly with the project’s long-horizon scenario requirements.

3 The Role and Advantages of LLMs in This Framework

Within this framework, Large Language Models do not serve as speculative stock price predictors, but rather act as an advanced **Semantic Parsing and Information Retrieval (IR) Layer** that bridges the investor’s unstructured personal thesis with structured quantitative portfolio constraints.

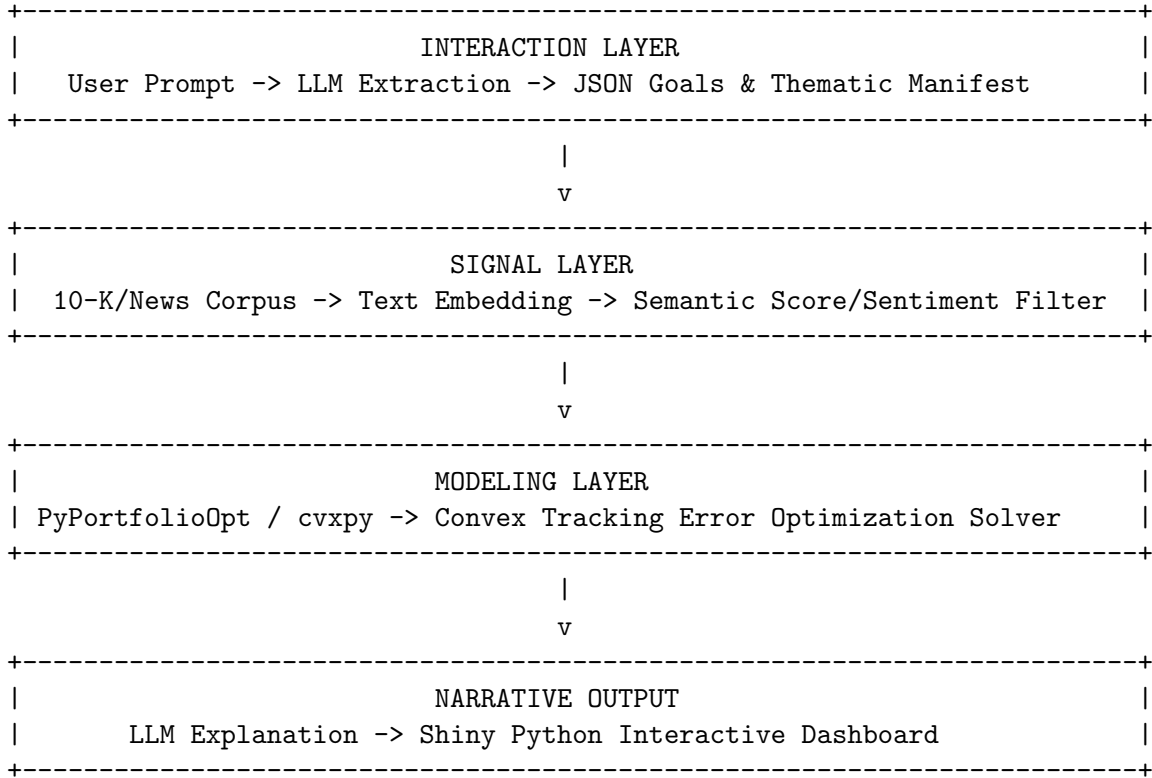
- **Unstructured Thematic Interpretation:** Investors rarely express their values in standard industry classification codes (like GICS). They express them as qualitative narratives: *“I want to invest in companies driving green hydrogen technology, but I want to avoid any company with recent labor disputes.”* LLMs excel at parsing these unstructured blocks of text, executing Named Entity Recognition (NER), and mapping these qualitative desires into a structured set of thematic keywords and sentiment scores.
- **Automated Universe Construction via Document Mining:** Replicating the core concept of the FINDALL architecture proposed by Schroeder and Kronseder (2024), LLMs

can be utilized to scan massive corpora of non-structural financial data — including corporate 10-K/10-Q filings, earnings call transcripts, and global financial news. By converting these documents into dense vector embeddings and performing semantic search or similarity matching, the LLM assigns a “Thematic Affinity Score” to every constituent stock in the benchmark index, programmatically constructing a customized investable universe.

- **Automated, Explainable Narrative Generation:** When a quantitative model shifts asset weights (e.g., trimming Apple and adding Nvidia), traditional systems only output numerical deltas. For individual clients, this is alienating. LLMs can ingest the optimization outputs (changes in asset weights and tracking error budgets) alongside the real-world news context, and automatically generate a compelling, natural-language narrative: *“We have increased your exposure to Company X because its recent patent filing aligns heavily with your custom ‘Next-Gen Solid State Battery’ thesis, while maintaining your portfolio’s tracking error tightly within your 1.5% risk tolerance budget.”* This directly fulfills the professor’s rigid requirement of using LLMs to manage narratives to impress audiences.

4 Integrated Framework Design and Implementation Approach

The overall framework is designed as a modular, three-layer architecture optimized for execution by a 3-person team, feeding into a unified Shiny Python Dashboard interface.



4.1 Layered Architecture Specifications

1. **The Interaction Layer (LLM-Driven):** This layer manages client onboarding via a conversational interface. The LLM conducts a multi-turn dialogue to extract the user’s base financial constraints (investment horizon T , core benchmark preference, risk aversion parameter) and their unique thematic choices. The output is strictly formatted using **JSON Schema** constraints to ensure deterministic pipeline passing.

2. **The Signal Layer (Advanced NLP & Information Retrieval):** This layer handles the 80% heavy-lifting LLM engineering work. We will pre-process a textual database of corporate disclosures and financial news. Using an open-source embedding model, we compute vector similarities between the client’s customized thematic manifest and each company’s business description. The final output of this layer is a continuous **Thematic Relevance Index** (S_i) for every asset i in the benchmark universe.
3. **The Modeling Layer (Core Quantitative Engine):** Upon receiving the asset weights of the benchmark index (e.g., *S&P* 500 weights w_b) and the historical asset covariance matrix Σ (derived via *yfinance*), this layer solves a standard **Constrained Tracking Error Minimization** problem:

$$\begin{aligned}
& \min_w && (w - w_b)^T \Sigma (w - w_b) \\
& \text{subject to} && \sum_i w_i = 1 \\
& && w_i \geq 0, \quad \forall i \\
& && \sum_i w_i S_i \geq S_{\text{target}}
\end{aligned}$$

where S_{target} is the minimum required thematic exposure requested by the investor. By framing this as a convex quadratic program, we can solve it in milliseconds using *PyPortfolioOpt* or *cvxpy*.

4.2 Dashboard Integration (Shiny Python)

The project will be deployed as an independent Tab within the interactive Shiny Python environment, broken into four sub-displays:

- **Sub-tab 1 (Interactive Chatbot Prompting):** Real-time visualization of the LLM parsing user text into structured JSON themes.
- **Sub-tab 2 (Thematic Universe Explorer):** A visual scatter plot showing which stocks have the highest semantic affinity scores and why they were selected.
- **Sub-tab 3 (Portfolio Weights & Tracking Metrics):** Interactive charts (via Plotly) comparing the custom Direct Indexing weights against the benchmark and tracking error statistics.
- **Sub-tab 4 (Narrative Attribution Report):** A text pane containing the LLM-generated narrative explaining the rationale behind the portfolio allocation.

5 Anticipated Challenges and Mitigation Strategies

1. **Dimensionality of Single-Stock Optimization:** Optimizing over all 500 components of the *S&P* 500 can scale up computing times, potentially slowing down the Shiny app interface.
Mitigation Strategy: We will implement a two-stage filtering process. The Signal Layer will first use the LLM-derived thematic score to filter down the index to the top N (e.g., 50 to 100) most relevant assets that also maximize industry diversification.
2. **Inconsistency and Hallucination in Thematic Matching:** LLMs might occasionally misclassify a company due to semantic ambiguity or corporate greenwashing in 10-K texts.
Mitigation Strategy: We will enforce rigid deterministic guardrails. The LLM’s semantic matching outputs will be cross-referenced with a hardcoded, verified taxonomy (such as GICS sector classifications). Any asset flagged for inclusion must pass an automated validation check.

3. **Covariance Matrix Instability and Optimization Failures:** Historical asset covariance matrices can occasionally become non-positive definite due to missing data or extreme collinearity, causing portfolio optimization packages to throw runtime errors.

Mitigation Strategy: We will utilize the Ledoit-Wolf shrinkage estimator provided natively inside PyPortfolioOpt (`risk_models.CovarianceShrinkage`) instead of the sample covariance matrix. This mathematically guarantees that the matrix is well-conditioned and positive definite.

4. **Engineering Asynchronicity and Team Constraints:** Managing asynchronous LLM API requests while maintaining a smooth, reactive UI in a small 3-person team can lead to integration gridlocks.

Mitigation Strategy: The project architecture enforces complete decoupling. The LLM text processing layer will operate as an independent preprocessing pipeline that dumps its data into lightweight SQLite databases or JSON files, reducing coupling.

5. **Demonstrating Model Persuasiveness:** To convince the academic advisors of the model’s practical superiority, we must provide robust empirics.

Mitigation Strategy: We will run extensive historical backtests across distinct market regimes (e.g., the 2022 inflationary bear market vs. the 2023–2024 AI bull market). We will measure and report the **Information Ratio**, **Active Share**, and **Thematic Fidelity Tracking** to prove its edge.